PROJECT DOCUMENTATION

Kolkata's Decadal Population & Water Infrastructure Forecast Dashboard

An interactive forecasting tool combining Civil Engineering design norms with Machine Learning models to assess Kolkata's urban water infrastructure

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DOMAIN	Civil Engineering × Data Science
TECH STACK	Python | Streamlit | Scikit-learn | Plotly
DATA SOURCE	Indian Census — Decadal Population Records
FOCUS CITY	KOLKATA [Kolkata Municipal Corporation (KMC)]
YEAR	2026

Population Forecast → Water Demand → Wastewater Generation → STP Capacity Analysis

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1. Executive Summary

This project is an interactive web dashboard built with Python and Streamlit that forecasts Kolkata's future population and then translates those forecasts into concrete civil engineering outputs — water demand, wastewater generation, sewer flow rates, pollution loads, and sewage treatment plant (STP) capacity utilisation.

The dashboard bridges two disciplines:

- Data Science — using both classical statistical methods and machine learning models to predict population growth across decadal intervals.
- Urban Water Engineering — applying standard IS/ CPHEEO design norms to convert population figures into infrastructure metrics.

The primary audience includes urban planners, civil engineers, environmental consultants, and students of environmental engineering who need a quick, reproducible tool to assess whether Kolkata's existing and planned sewerage infrastructure can keep pace with its growing population.

2. Problem Statement

Kolkata is one of India's densest and fastest-growing metropolitan areas. Adequate water supply and effective wastewater disposal are critical public health concerns. The key questions this project addresses are:

- How large will Kolkata's population be in a chosen future year?
- What volume of drinking water will need to be supplied each day?
- How much wastewater will be generated and what pollutant loads does it carry?
- What peak and average flows must the sewer network handle?
- Can the existing network of STPs handle this load, and how much headroom (or deficit) exists even after planned new plants come online?

3. Data Source

The dataset used is kolkata_population.csv, derived from Indian Census records. It contains decadal census data (Year, Population) for Kolkata Municipal Corporation (KMC).

Field	Type	Description
Year	Integer	Census year (decadal: 1901, 1911, … 2011)

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Field	Type	Description
Population	Integer	Recorded population of Kolkata for that census year

Only two columns are required. The code strips whitespace from column names before use to handle any formatting inconsistencies in the CSV.

4. Technology Stack

Library / Tool	Version	Purpose
Python	3.10+	Core programming language
Streamlit	1.x	Interactive web application framework
Pandas	2.x	Data loading, cleaning, and manipulation
NumPy	1.x	Numerical computations and array operations
Scikit-learn	1.x	Linear & Polynomial Regression ML models
Plotly (Graph Objects + Express)	5.x	Interactive data visualisation charts
Streamlit Caching	Built-in	@st.cache_data and @st.cache_resource for performance

5. Application Architecture

The application is structured as a single-page Streamlit app with the following logical flow:

Layer	Component	Responsibility
Data Layer	load_data()	Reads and caches kolkata_population.csv
Model Layer	train_models()	Trains & caches Linear + Polynomial Regression models
Forecast Layer	Arithmetic / Geometric / ML methods	Generates population predictions for the target year
Engineering Layer	Water Demand, WW, Sewer, STP modules	Converts population to infrastructure metrics
Presentation Layer	Streamlit tabs, metrics, Plotly charts	Renders all results interactively in the browser

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All heavy computations (data loading and model training) are wrapped in Streamlit's caching decorators (@st.cache_data and @st.cache_resource) to ensure the app remains fast across user interactions.

6. Population Forecasting Methods

Four independent methods are used in parallel, enabling direct comparison of their predictions:

6.1 Arithmetic Increase Method

A classical civil engineering method. The average absolute increase per decade is calculated from historical data and extrapolated linearly.

Formula:

$$P_n = P_0 + n \times \Delta P_{avg}$$

Symbol	Meaning
P_n	Forecast population at target year
P_0	Last known census population (base year)
n	Number of decades from base year to target year
ΔP_{avg}	Mean of decadal population increments (np.diff)

Best used when: Growth is roughly constant in absolute numbers.

6.2 Geometric (Percentage) Increase Method

Models growth as a compounding percentage. The average growth rate per decade is calculated and applied as compound interest.

Formula:

$$P_n = P_0 \times (1 + r)^n$$

Symbol	Meaning
r	Mean of decadal percentage growth rates
n	Number of decades from base year to target year
P_0, P_n	Base and forecast populations

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Best used when: Growth follows an exponential or multiplicative pattern (common in rapidly developing cities).

6.3 Linear Regression (Machine Learning)

Scikit-learn's LinearRegression model is trained on all historical (Year, Population) pairs. The year is the independent variable; population is the target. Predictions are made directly from the model.

Key implementation details:

- Year array is reshaped to a 2-D column vector: years.reshape(-1, 1)
- The model is cached with @st.cache_resource to avoid retraining on every slider move.
- Produces an optimal straight-line fit minimising the sum of squared residuals across all census years.

Note: Population estimated by Linear Regression is used as the basis for all downstream water infrastructure calculations.

6.4 Polynomial Regression (Machine Learning)

Extends Linear Regression by introducing quadratic terms. PolynomialFeatures(degree=2) from Scikit-learn transforms the year into [1, year, year^2]. A LinearRegression model is then fit on this expanded feature set.

- Captures non-linear (accelerating or decelerating) growth trends.
- Can overfit for very distant future years; use with caution for multi-decade extrapolation.
- Same caching strategy as Linear Regression.

7. User Interface & Controls

UI Element	Location	Function
Year Input (number_input)	Sidebar	User selects any future year (2021–10000) in steps of 10
Forecast Tab	Main — Tab 1	Displays four metric cards with population predictions
Visualization Tab	Main — Tab 2	Plotly line chart comparing all four forecast curves vs. historical data
Dataset Tab	Main — Tab 3	Displays the raw CSV as a scrollable dataframe

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UI Element	Location	Function
Engineering Sections	Below tabs	Metrics and bar charts for all infrastructure calculations
Engineering Assumptions	Expander	Documents all design constants used in calculations
Plant Capacity List	Expander	Lists all existing and future STP capacities in MLD

8. Water & Wastewater Engineering Calculations

All engineering calculations use the Linear Regression population forecast as the design population. This section documents every formula and assumption applied.

8.1 Water Demand Estimation

Per capita daily water supply norm: 135 litres per capita per day (lpcd), as per IS 1172 / CPHEEO guidelines.

Formula:

Water Demand (litres/day) = Population × 135

Water Demand (MLD) = Water Demand (litres/day) ÷ 1,000,000

8.2 Wastewater Generation

Standard practice assumes 80% of water supplied is returned as wastewater (accounting for evaporation, industrial use, and non-return losses).

Formula:

Wastewater (litres/day) = Water Demand × 0.80

8.3 Wastewater Quality Analysis — Pollutant Loads

Parameter	Concentration	Formula	Unit
BOD (Biochemical Oxygen Demand)	250 mg/L	BOD Load = Wastewater (L/day) × 250 ÷ 1,000,000	kg/day
COD (Chemical Oxygen Demand)	500 mg/L	COD Load = Wastewater (L/day) × 500 ÷ 1,000,000	kg/day

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These concentrations represent typical domestic sewage quality per CPHEEO Manual on Sewerage and Sewage Treatment.

8.4 Sewer Flow Model

Three sewer flow design values are computed from the wastewater flow:

Flow Type	Factor	Formula	Use in Design
Average Flow	1.0×	Equal to wastewater generation	Baseline hydraulic load
Minimum Flow	0.5×	50% of average	Self-cleansing velocity check
Peak Flow	2.5×	250% of average	Sewer pipe & pump sizing

8.5 Treatment Plant Load Distribution

Wastewater is distributed proportionally across STPs based on each plant's design capacity relative to total system capacity.

Formula:

Plant Load (MLD) = Total Wastewater (MLD) × (Plant Capacity / Total System Capacity)

Two scenarios are evaluated:

- Current infrastructure: 10 existing STPs with a combined capacity of 400 MLD.
- Future infrastructure: 10 existing + 6 planned STPs with a combined capacity of 753.5 MLD.

9. Sewage Treatment Plant Inventory

9.1 Existing Plants

STP Name	Capacity (MLD)
Garden Reach	57
Keorapukur	45
Bangur	45

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STP Name	Capacity (MLD)
Dhapa	80
Jorabagan	45
Watgunge	35
New Town (AA-IIC)	31
New Town (AA-IIB)	14
South Suburban East	30
Bagjola	18
TOTAL	400

9.2 Planned / Future Plants

STP Name	Capacity (MLD)
Garden Reach Extension	113.6
Palta Extension	90.9
Hossainpur STP	41.0
Joka (Bank Plot)	40.0
Joka (WBSETCL Campus)	45.0
Rania STP	23.0
Existing Plants Combined	400.0
TOTAL (after expansion)	753.5

10. Infrastructure Capacity Analysis

The dashboard computes a capacity utilisation percentage and gap for both the current and future STP networks:

Metric	Formula
Load % (Current)	(Wastewater MLD / 400 MLD) × 100
Load % (Future)	(Wastewater MLD / 753.5 MLD) × 100
Gap (MLD)	Wastewater MLD − System Capacity MLD (positive = deficit, negative = surplus)

Status indicators use Streamlit alerts:

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- Green (st.success): System load < 80% — infrastructure is sufficient.
- Orange (st.warning): System load 80–100% — nearing capacity.
- Red (st.error): System load > 100% — infrastructure overloaded.

11. Engineering Assumptions Summary

Parameter	Value	Standard / Source
Per capita water supply	135 lpcd	IS 1172 / CPHEEO
Wastewater return factor	80% of water demand	CPHEEO Manual
BOD concentration	250 mg/L	Typical domestic sewage
COD concentration	500 mg/L	Typical domestic sewage
Minimum flow factor	0.5× average	Standard sewer design
Peak flow factor	2.5× average	Standard sewer design
Forecast model for engineering	Linear Regression	Project design choice
Data interval	Decadal (10 years)	Indian Census cadence

12. End-to-End Output Pipeline

The following chain summarises the complete data and computation flow:

1. Census CSV loaded → Year & Population arrays extracted.
2. Four population forecasting models trained or calculated.
3. Target year selected by user via sidebar input.
4. Population predicted for target year by all four methods.
5. Linear Regression prediction used as design population.
6. Water Demand (MLD) = Design Population × 135 lpcd ÷ 10^6.
7. Wastewater (MLD) = Water Demand × 0.80.
8. BOD Load (kg/day) and COD Load (kg/day) computed.
9. Average, Minimum, and Peak Sewer Flows computed.
10. Wastewater distributed proportionally across existing and future STPs.
11. Capacity utilisation (%) and surplus/deficit (MLD) displayed with colour-coded status alerts.

13. How to Run the Project

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Prerequisites

- Python 3.10 or higher
- pip package manager

Installation & Launch

- Clone or download the project folder.
- Install dependencies: pip install streamlit pandas numpy scikit-learn plotly
- Ensure kolkata_population.csv is in the same folder as the app file.
- Launch the dashboard: streamlit run app.py

The app will open in your default web browser at http://localhost:8501.

14. Limitations & Future Scope

Current Limitations

- The polynomial regression model may produce unreliable forecasts for very distant future years due to overfitting tendencies.
- All engineering calculations use fixed national design norms; local variations in water use patterns are not modelled.
- East Kolkata Wetlands (EKW) natural treatment capacity is explicitly excluded; only STPs are considered.
- The dashboard does not account for climate change impacts on water demand.

Future Scope

- Integrate LSTM or ARIMA time-series models for improved long-horizon population forecasting.
- Add interactive per capita rate slider so users can test different consumption scenarios.
- Include EKW natural treatment capacity as a parallel scenario.
- Geo-visualisation overlay showing STP locations on a Kolkata map.
- Export report generation (PDF/CSV download buttons).

15. References

- Indian Census Data — Office of the Registrar General & Census Commissioner, India.
- CPHEEO Manual on Sewerage and Sewage Treatment Systems, 2013 — Ministry of Housing & Urban Affairs, Government of India.

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- IS 1172:1993 — Code of Basic Requirements for Water Supply, Drainage and Sanitation.
- Scikit-learn Documentation — sklearn.linear_model.LinearRegression, sklearn.preprocessing.PolynomialFeatures.
- Streamlit Documentation — https://docs.streamlit.io
- Plotly Python Graphing Library — https://plotly.com/python

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